

CCNA 200-301

Chapter 27 - Static NAT Configuration (Lab)

Introduction

Static NAT is a one-to-one mapping between a Private IP address and a Public IP address. It is commonly used for servers or devices that must always appear on the Internet with the same public address.

Lab Objectives

- Configure routing before NAT.
- Configure a Default Route toward the ISP.
- Configure Static NAT.
- Mark Inside and Outside interfaces.
- Verify connectivity and NAT translation.

Lab Topology Summary

Router0 represents the customer network. Router1, Router2 and Router3 represent the ISP. Server0 (12.0.0.2) and Server1 (14.0.0.2) simulate public servers.

Network Preparation

1. Configure OSPF between ISP routers (Router1, Router2 and Router3).
2. Configure a Static Default Route on Router0 toward Router1.
3. Ensure Router2 owns 12.0.0.1 and Router3 owns 14.0.0.1, then advertise both networks in OSPF.

Why NAT Was Needed

Without NAT, packets leave the internal network using the private source address 192.168.1.2. The external network cannot communicate with private addresses over the Internet, so replies fail. Static NAT replaces the private address with a reachable public address.

Static NAT Configuration

```
ip nat inside source static 192.168.1.2 10.0.0.1

interface fa0/0

    ip nat inside

interface fa0/1
```

```
ip nat outside
```

Verification

- Ping Server0 (12.0.0.2) from PC0.
- Use 'show ip nat translations'.
- Use 'show ip nat statistics'.
- Verify routing using 'show ip route'.

Troubleshooting (Real Lab Case)

During the lab, connectivity failed before NAT because Router2 and Router3 were not configured with IP addresses on the server-facing interfaces and the 12.0.0.0/24 and 14.0.0.0/24 networks were not advertised into OSPF. After assigning 12.0.0.1 and 14.0.0.1 to the router interfaces and adding the corresponding OSPF network statements, end-to-end connectivity succeeded.

Exam Notes ★

- Static NAT = One Private IP ↔ One Public IP.
- Each additional internal host requires another Public IP.
- Inside and Outside interfaces must be identified.
- Routing must work before NAT can be tested.
- Always verify with show ip nat translations.